Automated Process-Fluid Flow Skid

Lam Research Challenge 3.0 · Stage 2 Hardware Hustle · Team [Team name]

M
M · Mechanical
Hydraulics and build

E1
E1 · Electrical
Power and wiring

E2
E2 · Electrical
Control and Case 1

E3
E3 · Electrical
Case 2 and results

Set a flow, measure it, protect it, flush it

M

Case 1

Control and measurement

Circulate process fluid TK-01 → sump → TK-01 with P-01 at 50 % and P-02 at 100 %. Measure and totalise the return flow. If fluid is suddenly dumped into the sump, stop P-01, keep P-02 draining, then resume.

Case 2

Flush and shutdown

Isolate TK-01 with the solenoid, flush from the 4.2 L tank with all three pumps running, then switch everything off with the button once the flush water is gone.

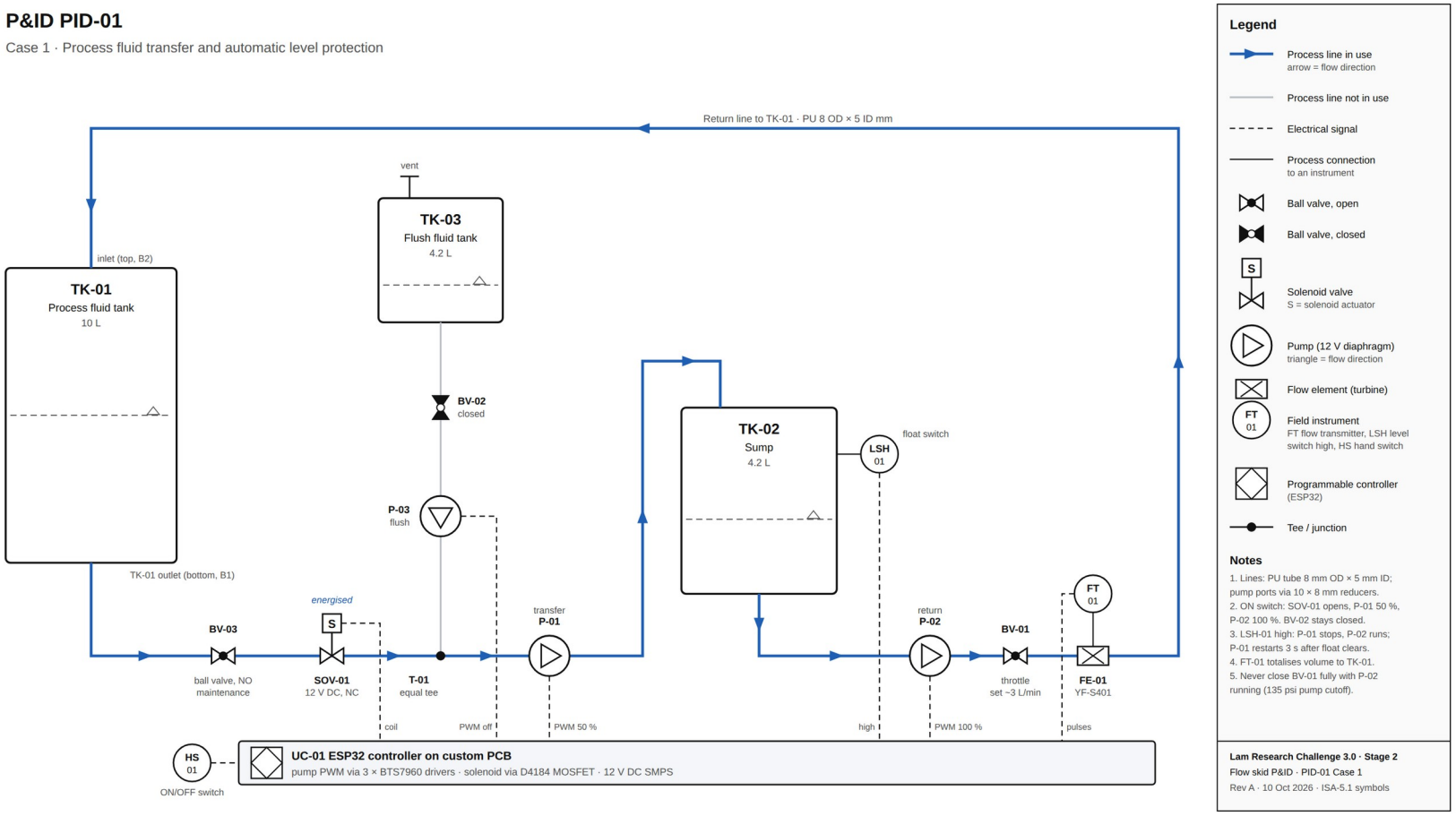
8 mm
pipe diameter (paper)

1–6
L/min required range

3 L/min
nominal target flow

0
hands on valves during the run

Case 1 P&ID: follow the blue line



TK-01 → BV-03 → SOV-01 → T-01 → P-01 → TK-02 sump → P-02 → BV-01 → FT-01 → back to TK-01

Every component, in flow order

M

Case 1 loop

- **TK-01** — **10 L process tank**
  bottom outlet, top return
- **BV-03** — **ball valve**
  maintenance shut-off, open
- **SOV-01** — **12 V solenoid**
  isolates TK-01 in Case 2
- **T-01** — **equal tee**
  flush line joins here
- **P-01** — **transfer pump**
  50 % PWM into sump
- **TK-02** — **4.2 L sump**
  top inlet, bottom outlet
- **P-02** — **return pump**
  100 % PWM
- **BV-01** — **ball valve**
  throttle, sets the flow

Flush branch

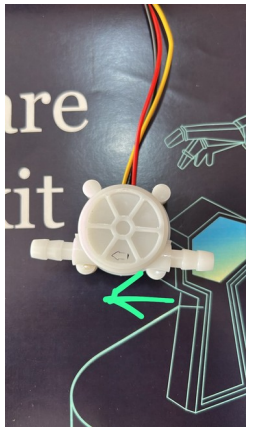
- **TK-03** — **4.2 L flush tank**
  bottom outlet, vented top
- **BV-02** — **ball valve**
  left open; idle P-03 holds back
- **P-03** — **flush pump**
  feeds the tee

Instruments

- **FT-01** — **flow transmitter**
  YF-S401 turbine, pulses
- **LSH-01** — **float switch**
  sump high level
- **HS-01** — **ON/OFF button**
  starts the sequence



Pump



FT-01



SOV-01

Insert screenshot
Labelled photo of our finished skid

Built for the tube we actually have

Paper says 8 mm. The kit tube is 8 mm OD × 5 mm ID.

at 3 L/min	8 mm bore	5 mm bore
Velocity	0.99 m/s	2.55 m/s
Reynolds no.	7 960	12 700
Friction loss	≈ 2 kPa/m	≈ 19 kPa/m

Same flow, 2.56 × faster water and about 9 × the friction. That is why our loop delivers about 2 to 2.5 L/min, not the pump's 6.1 L/min free-flow rating.

1. **Joints**
   Brass bulkhead with a washer each side at every tank port; Teflon tape on every thread.

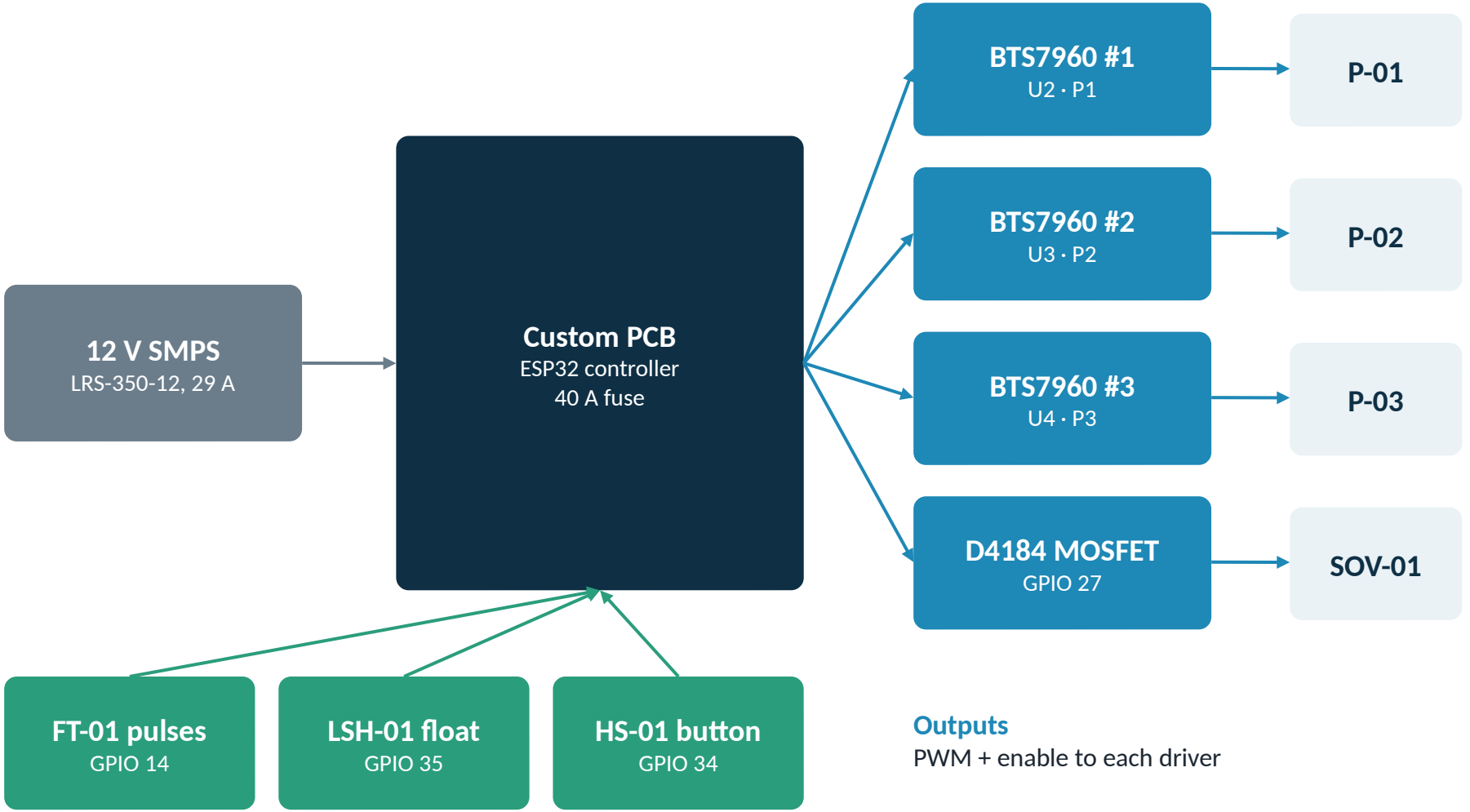
2. **Pump ports**
   Short 10 × 8 mm blue tube with reducers only at the pumps; 5 mm PU tube elsewhere.

3. **Leak test**
   Water in, pumps off, then running, before any electronics came near the wet side.

4. **Safe limits**
   Never close BV-01 with P-02 running (135 psi cutoff). Start TK-01 at about 5 L so the flush cannot overflow it.

One 12 V supply, one ESP32, three drivers

E1



Outputs
PWM + enable to each driver



≈ 11 A

worst case in Case 2: three pumps at up to 3.5 A plus the solenoid

Three faults, one symptom: nothing moves

E1

P-02 driver dead

Cause
No R_PWM2 trace on the PCB (organiser notice).

Fix
Drive P-02 from L_PWM2, tie the driver's unused RPWM input high, invert in firmware.

FT-01 reads 0

Cause
Zero edges on GPIO 14: red and black swapped on S-V-G, so the Hall chip had reversed supply.

Fix
Red → V, black → G, yellow → S. Then mounted it vertically, flow upward, so it runs full; flat, it under-read.

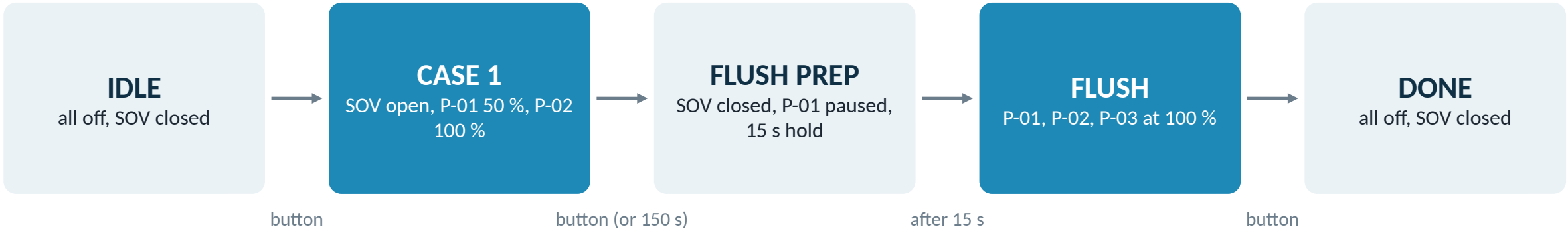
Button ignored

Cause
Shorting SW to GND worked, the switch did not.

Fix
Wired across diagonal legs of the tact switch.

The button steps through every case

E2



1 **Float high for 300 ms**
P-01 (and P-03 in the flush) stop; P-02 keeps draining the sump.

2 **Float clear for 3 s**
P-01 restarts. The delay stops a sloshing surface from chattering the pump.

3 **0.6 s soft start**
Every pump ramps up, so three motors never hit the SMPS at once.

4 **Fail-safe**
Reset or power loss lands in IDLE with the solenoid closed. Typing stop ends everything from any state.

From pulses to litres

E2

1

Pulse
FT-01 rotor turns; each Hall pulse fires an interrupt on GPIO 14

2

Filter
Edges closer than 300 µs rejected; 6 L/min is only ≈ 590 Hz

3

Frequency
every 0.5 s: $f = N / t$

4

Flow
$Q = a \cdot f + b$ (our calibration)
$mL/s = L/min \times 1000 / 60$

5

Total
litres returned = $\Sigma\, Q \cdot \Delta t$ streamed as CSV, plotted live

≈ 98 Hz
per L/min: datasheet start value, replaced by our fit

50 mL/s
equals the 3 L/min target

0.5 s
logging period, one CSV row per sample

Case 1 live: sudden charge into the sump

E2

1. **Press ON**
   SOV-01 clicks open, P-01 50 %, P-02 100 %
2. **Steady flow**
   FT-01 ≈ [__] L/min = [__] mL/s, total climbing
3. **Pour ≈ 1 L into sump**
   LSH-01 trips: P-01 stops, P-02 keeps running
4. **Level falls**
   float clear 3 s, P-01 restarts, loop resumes

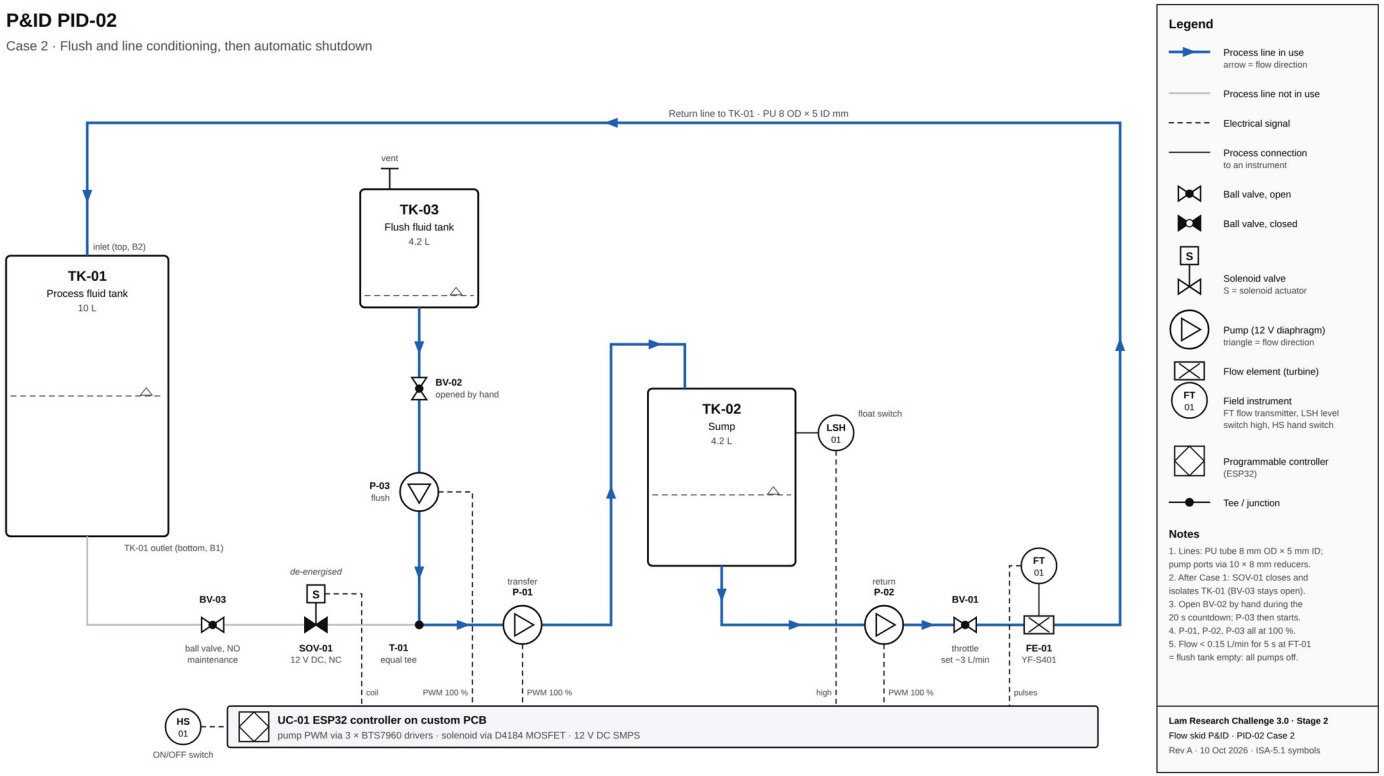
Insert screenshot
Flow vs time in mL/s from flow_logger.py, with the sump-charge event marked

[___] L
total returned to TK-01

[___] L/min
steady return flow

Case 2 live: flush, then stop on the button

E3



TK-03 → BV-02 → P-03 → T-01 → P-01 → sump → P-02 → FT-01 → TK-01

Stop

When the flush tank is empty and FT-01 falls toward zero, one press of the button switches every pump off. SOV-01 stays closed.

Protection

The float switch still stops P-01 and P-03 if the sump runs high during the flush.

Check

Flush total [__] L vs 4.2 L put in.

We calibrated FT-01 instead of trusting it

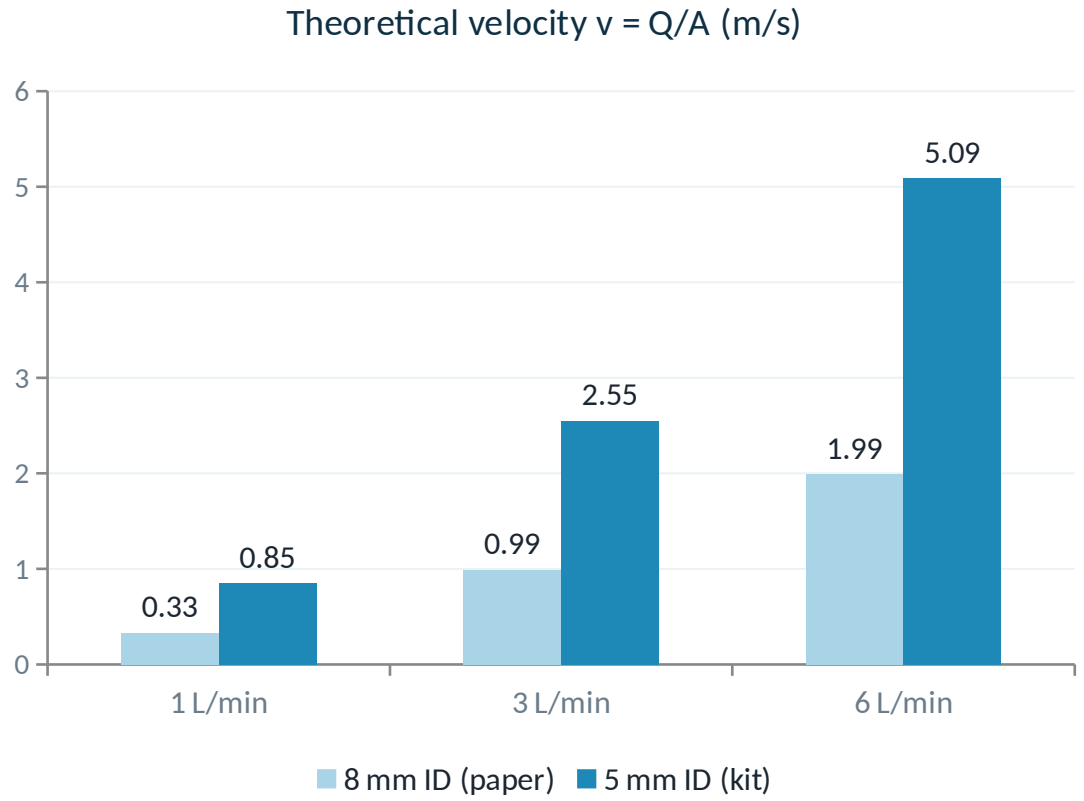
E3

Trial	N	t (s)	f (Hz)	V (L)	Q (L/min)
1					
2					
3					
4					
5					

Q = a·f + b

[__]	[__]	[__]	[__] %	[__] %
a	b	R^2	max err	avg err

Five timed bucket trials: f = N/t, Q = V/t, least-squares fit.



Measured: the loop tops out near [2–2.5] L/min, so 6 L/min is out of reach with a 5 mm bore, BV-01 and the FT-01 nozzle.

A ball valve is a coarse throttle

E3

Insert screenshot

Q2.1 graph: BV-01 position (%) vs flow (L/min) at 25, 50, 75, 100 %

1

Quick opening

Most of the flow arrives in the first half of the quarter turn.

2

Flat at the top

Above 75 % the pump, tubing and FT-01 set the flow, not the valve.

3

Our decision

Hold 3 L/min by trimming BV-01 and P-02 PWM, not by valve angle alone.

One button. Measured, protected, flushed.

Case 1 and Case 2 in one run, stepped by the ON/OFF button, with no hand on a valve.



Thank you